Apex Precision Components - Strategic Recommendations

Problem Statement –

Aligning two distinct business tech operations models based on on-premises based v/s cloud-based applications and database management tools by July 2026 timeline, covering following criterion:

1. Address concerns over cloud-based technologies regarding application and data security and reliability
2. Predict cost and ROI with impact on financials over long-term
3. To improve delivery commitment timelines post-merger

Three Alternatives and Relative Business Impact of Such Decisions:

Options	Business Impact
Migrating SLS to On-prem - Moving all SLS GC applications, database and warehouse to Apex's datacenter	1. ~$800,000 upfront expansion cost. 2. ~18-month timeline—risking July 2026 commitments. 3. APC's data center has limited room for growth and high DR cost. 4. Requires SLS team to learn on-prem systems. 5. Forces SLS customers onto a retrograde setup, reducing platform value and harming acquisition credibility.
Hybrid Integration - Keep both environments and build integration pipelines	1. Minimum ~$540,000 annual combined budget. 2. New security tools and skills needed in both teams. 3. Higher complexity and longer troubleshooting cycles. 4. Dual security models create overhead to protect customer-specific data. 5. APC's existing $180,000 maintenance cost continues and will grow.
Migrating only non-manufacturing APC systems to cloud	1. Upskills APC team and increases productivity. 2. Cross-skill alignment with SLS's cloud-native team. 3. Shifts APC from capital expenses to pay-as-you-go cloud budgeting. 4. Faster time to market and improved security posture through modern cloud services. 5. Higher availability, automatic scaling, and improved DR without additional physical expansion.

Recommendation Using AWS CAF's Foundational Capabilities

Business Perspective

Cloud adoption reduces overall tech cost, increases time-to-market speed, and removes the need for data-center expansion. AWS benchmarks show 27% user-level cost reduction and meaningful savings from retiring on-prem maintenance ($180,000 annually). This path supports the acquisition's revenue goal by allowing fast integration by July 2026 - critical commitments made. Outcome: better ROI visibility, reduced capital spending, and stronger competitive position.

People Perspective

Cloud migration frees APC's IT team from infrastructure maintenance and shifts them toward higher-value work. Cross-training APC and SLS teams builds shared capability in cloud, DevOps, and manufacturing systems. The change also supports flexible ways of working across engineering, operations, and analytics teams.

Governance Perspective

A cloud-based consumption model provides predictable spend tracking, automated reporting, and better alignment with finance's need for cost transparency. Standardized policies across environments also reduce audit effort and support compliance with customer and regulatory requirements.

Platform Perspective

Cloud services simplify APC's current environment by moving ERP, analytics, and office systems off legacy servers. Managed services (databases, analytics tools, serverless compute) allow IT to focus on application needs instead of hardware upkeep. Hybrid support (VPN, DirectConnect, VMware options) ensures real-time manufacturing workloads can remain on-prem without blocking modernization.

Expected gains: less downtime, scalable analytics, and smoother integration with SLS systems.

Security Perspective

AWS offers multi-layer access control, encryption, network isolation, and the option to manage encryption keys—critical for APC's engineering files and ITAR-related obligations. CloudTrail, IAM, WAF, and Inspector improve visibility and monitoring compared to APC's current on-prem model. This path strengthens APC's security posture beyond what is possible with limited physical expansion at the Columbus data center.

Operations Perspective

Cloud migration increases system resilience, enables automatic scaling, and reduces downtime from the current 99.7% baseline toward near-continuous availability. APC avoids further DR expansion costs and gains faster recovery options; each avoided hour of downtime saves ~$12,000. Operational teams spend less time on patching, hardware lifecycle issues, and storage management, improving overall productivity.

Final Recommendation

APC should adopt a cloud-migration strategy (non-manufacturing systems on cloud & shop-floor systems on-premises). This approach fits AWS CAF guidance across business value, cost control, people readiness, governance, platform modernization, security, and operational resilience. This strategy preserves SLS's strengths, avoids unnecessary re-architecture, prepares APC for scalable analytics capabilities, and reduces long-term infrastructure risk, thus positioning for reliable integration by July 2026 and supporting future competitiveness.